

	Answer	Mark
5(a)(i)	The only correct answer is C. <i>A is incorrect because m is an area so two dimensional</i> <i>B is incorrect because year is one dimensional</i> <i>D is incorrect because year is one dimensional</i>	(1)

	Answer	Mark
5(a)(ii)	The only correct answer is B. <i>A is incorrect because $NPP = GPP - R$</i> <i>C is incorrect because $NPP = GPP - R$</i> <i>D is incorrect because $NPP = GPP - R$</i>	(1)

Question number	Answer	Additional guidance	Mark
5(a)(iii)	98 : 10 : 1	ACCEPT 98 : 9.6 : 1 98.3 : 9.6 : 1 100 : 10 : 1 10 : 1 : 0.1 10.3 : 1 : 0.1 1 : 0.1 : 0.01	(1)

Question number	Answer	Additional guidance	Mark
5(a)(iv)	An explanation that includes the following points: • because not enough energy (in trophic level 3) (1) • to sustain (organisms in) a fourth trophic level (1) OR • energy is lost between trophic levels (1) • so not enough energy for another trophic level (1)	IGNORE no energy / biomass ACCEPT support IGNORE no energy / biomass	(2)
Question number	Answer	Additional guidance	Mark
5(a)(v)	A description that includes the following points: • breakdown organic matter (1) • with (hydrolytic) enzymes (1) • releasing carbon <u>dioxide</u> from respiration (to the atmosphere) (1) • where it is used for photosynthesis (by plants) (1)	ACCEPT dead {tree / gerenuk / lion} / tissues / named organic molecule ACCEPT named enzyme NB release {digestive / hydrolytic} enzymes onto dead organisms = 2 marks	(3)

Question number	Answer	Additional guidance	M
(b)	Level 1 points: • due to natural selection (Descriptions of differences in features) • gerenuk's food is higher up / springbok's food is lower down • gerenuks have longer {necks / limbs} • gerenuk's white areas are {underneath / more concealed} • better adapted organisms survive and reproduce • passing their alleles onto their offspring Level 2 points: • occupy a different niche because they eat different food (Descriptions of how mutation caused a difference) • change in length of {neck / limbs} due to a mutation • change in type of horns due to a mutation (Explanations of how a features are an adaptation / result in difference in behaviour) • organisms with longer neck could reach the higher leaves • organisms with smaller horns did not get caught up in branches • smaller organisms had to go to waterhole for water • organisms with larger horns could defend off predators • male gerenuks kept horns for fighting for females • reference to {reduced gene flow / change in allele frequency / reproductive isolation} with no links made Level 3 points: • length of {neck / limbs} is a polygenic characteristic • type of horns is a polygenic characteristic • therefore organisms will show (continuous) variation for this phenotype • height of food acted as a selection pressure on size • {branches / predators} acted as a selection pressure on horns • organisms that {occupy a different niche / feed on different food} are not in competition with each other • two groups of organisms moved apart depending on where their food was • sympatric speciation / organisms not separated by a physical barrier • therefore reduced gene flow (between the two groups) • resulting in changes in the allele frequency • eventually resulting in reproductively isolated • and the formation of two species	Level 1: 1 mark = 1 point made from any level 2 marks = 3 points made from any level Level 2: NB Must be a reference to the data / context of the question 3 marks = 4 points made which include at least one level 2 point 4 marks = 5 points made which include at least one level 2 point Level 3: NB Horns and {leg / neck} length must be included 5 marks = 6 points made which includes at least one level 3 point 6 marks = 7 points made which includes at least one level 3 point	(